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Cell wall distraction and biofilm inhibition of marine *Streptomyces* derived angucycline in methicillin resistant *Staphylococcus aureus*

Thankaraj Rajam Jabila Mary [a], Rajaretinam Rajesh Kannan [a,*], Appadurai Muthamil Iniyan [b], Dasnamoorthy Ramachandran [a], Samuel Gnana Prakash Vincent [b]

[a] *Infectious Disease Biology Lab, Centre for Molecular and Nanomedical Sciences, Sathyabama Institute of Science and Technology, Jeppiaar Nagar, Rajiv Gandhi Salai, Chennai, 600119, Tamil Nadu, India*

[b] *Centre for Marine Science and Technology (CMST), Manonmaniam Sundaranar University, Rajakkamangalam, Kanyakumari, 629502, Tamil Nadu, India*

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ABSTRACT

The emergence of life threatening antibiotic resistant pathogens and its associated mortality and morbidity necessitates many new antibiotics from diverse ecological habitats. Marine sponge associated microbes are promising to provide such antimicrobial compounds. In the present study, we report antibacterial and *anti*-biofilm potential of the angucycline antibiotic 8-*O*-metyltetrangomycin from *Streptomyces* sp. SBRK2 isolated from a marine sponge of Gulf of Mannar, Rameswaram, India. Our screening program to tackle methicillin-resistant *Staphylococcus aureus* (MRSA) drug resistance from marine sponge associated actinobacteria yielded the bioactive strain SBRK2. Based on 16S rRNA gene phylogenetic analysis the isolate was found to closely related with *Streptomyces longispororuber* NBRC 13488^{T}. *In vitro* production by agar plate fermentation, solvent based extraction, TLC, HPLC purification and LC–MS based de-replication revealed the bioactive compound as 8-*O*-metyltetrangomycin. The antibacterial minimum inhibitory concentrations against MRSA was identified as 2 μg/mL. Sub-inhibitory concentration of the compound 8-*O*-metyltetrangomycin reduced the biofilm formation of *S. aureus* ATCC25923 and increased the cell surface hydrophobicity index. Scanning electron microscopic observation of the sub-inhibitory concentration exposure revealed a wrinkled membrane surface and slight cellular damage shows the cell wall distracting property of the compound. Zebrafish embryo based toxicity assays exhibited 100 μg/mL of compound as maximal non-lethal concentration which had demonstrated the positive relationship in safety index. The angucycline compound 8-*O*-metyltetrangomycin could be a potential candidate for the development of *anti*-biofilm agents against drug resistant pathogens.

1. Introduction

Infectious diseases caused by multidrug-resistant bacteria (MDRB) are consequential universal threat. Especially in developing countries for centuries infectious disease is been among the leading causes of mortality, disorder, growing challenges to health security and human progress [1]. The aberrant usage of antibiotics is a major cause for the increasing number of antimicrobial-resistant pathogens. Therefore finding a novel effective target specific drug against multi-drug resistant bacteria is a need of the hour [2]. One among the antibiotic resistant disease causing bacteria is *S. aureus,* which is becoming a serious medical threat as it rise in their incidence and also resurgence towards antibiotic and more lethal [3]. This ubiquitous *S. aureus* has adapted to a broad array of host species, this includes human host, it survives as a natural flora on the surface of the normal human skin here it doesn't cause any disease usually. Competing rewards is treating MRSA infections has augment research on prophylaxis and treatment strategies. Indeed, several new drugs (Oritavancin, TD -1792, Iclaprim and Dalbavancin) are currently under different stages of development, including some which have already been approved (Telavancin, Ceftobiprole and Tedizolid) by the Food and Drug Administration (FDA) and the European Medicines Agency (EMA) [4].

Many pathogens form a biofilm as a resistance strategies which makes them indestructible than their planktonic counterparts [5]. Biofilms could be involved in over 60% of microbial infections [6,7] while two-thirds of all human bacterial infections are caused by the biofilms [8].Consequently, Biofilm-forming mutants of methicillin-resistant *S. aureus* (MRSA) infections are very difficult to cure [9]. The

* Corresponding author.
E-mail addresses: rajeshkannan.mnru@sathyabama.ac.in, rajeshnbt12@gmail.com (R.R. Kannan).

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cumulation of these problems along with few numbers of new antimicrobials grants stimulation to intensify the research for potential drugs and its derivatives from different sources [10]. This paved a way to focus on unexplored natural products, further research is propitious because a great percentage of many new approved antibacterial are either natural products themselves, or natural product derivatives [11].

Marine life forms are an important source of structurally diverse and biologically active secondary metabolites, several of which have inspired the development of new classes of therapeutic agents. Previously many compounds are isolated from macro organisms such as sponges, tunicates, etc, are actually metabolic products of associated microbes [12]. Thus, attention turned to smaller and smaller creatures such as marine actinomycetes, marine fungi, and diverse other groups of marine eubacteria. This emphasis on microorganisms has been rewarded with a wealth of new natural products [13]. In the present study, we aimed to isolate marine sponge associated actinobacteria for its potential to produce *anti*-MRSA active compounds and its *anti*-biofilm potential.

2. Materials and methods

2.1. Sample collection and isolation of actinomycetes

Marine sponge *Spirostella* sp. was collected from Pampan Island, Gulf of Mannar, Rameswaram Coast, Bay of Bengal, India using by-catch method in the fishing nets. Collected sponge was transferred to plastic bags containing sea water and transported to the laboratory. Sponge specimens were rinsed in sterile sea water, cut into pieces of approximately ~1 cm^3 thoroughly homogenized in a sterile mortar with 10 vol of sterile sea water. ISP 2 medium comprising 0.4% glucose, 0.4% yeast extract, 1% malt extract, 0.2% $CaCO_3$, 0.02% chloramphenicol, and 1.5% agar in 50% sterile water and Actinomycetes isolation medium (AIM) containing soluble starch 20 g, KNO_3 1 g, NaCl 0.5 g, K_2HPO_4 0.5 g, $MgSO_4$ 0.5 g, $FeSO_4$ 20 μM and agar 15 g in 1L were used for actinomycete isolation. Media were prepared in sea water and supplemented with aqueous sponge extract to enhance the growth of sponge specific symbiotic actinomycetes. Aqueous sponge extract was prepared by grinding 20 g of sponge biomass in a mortar containing 20 mL of sterile seawater followed by centrifugation (5000 rpm, 10 min) and sterilized by filtration through a 0.2 μm pore size filter [14]. Cycloheximide (100 μg/mL), nystatin (25 μg/mL) and nalidixic acid (25 μg/mL) were added in the sterilized media to inhibit fungal growth and fast-growing Gram-negative bacteria, respectively. After 3–7 days of incubation at 28 °C actinomycetes were picked based on their morphological characteristics and re-streaked several times to obtain pure colonies. The isolates were maintained on plates for short-term storage and long-term strain collections were preserved with 30% glycerol at −80 °C.

2.2. Anti-MRSA screening of isolated strains

Double layer agar method was employed to screen the *anti*-MRSA property of isolated strains [15]. The isolated actinomycete strains were patched on the centre of the Actinomycetes Isolation Media (AIM) plates and incubated at 28 °C for 5–7 days. After incubation, Luria broth soft agar with overnight culture of MRSA ATCC 33591 (5×10^4 CFU/mL) was prepared and overlaid on the culture medium separately. Plates were then incubated at 37 °C for 24 h and zone of inhibition was measured.

2.3. Characterization of active strain

2.3.1. Morphological characterization and identification

Morphological characteristics of isolated strains were examined by the visible observation of 14-day-old culture grown on AIM medium [16, 17]. Micromorphology, spore chain morphology and sporulation were observed under light microscope by cover slip culture method [18]. In this method, sterile cover slips were inserted aseptically into the sterile solidified agar medium in a Petri dish at an angle of 45°–60° by means of a sterile forceps and the strain was inoculated by streaking method, at the free space of the plate. The inoculated plate was then incubated at 28 °C for 7 days, 14 days and 21 days intervals. The hyphae of the strain grew and spread on the cover slip and produced spores. The cover slips were then taken out, mounted on slides and observed under microscope [19]. Colours of aerial and substrate mycelia were determined and recorded. For scanning electron microscope observation one set of coverslips were carefully placed the upper surface of circular aluminium stubs then coated, under vacuum, with a film of gold. The gold coating process was completed in 15–20 min. Once coated with gold, the specimens were ready for examination under the scanning electron microscope (FESEM-SUPRA 55, Carl Zeiss, Germany). The gold-coated metal stubs were viewed on the SEM at an accelerating voltage of 20 kV, a probe diameter of 102 pA, to obtain secondary electron images. The field was scanned at low magnification until the line of growth was detected. Areas with clear, intact spore structures were then selected for examination at higher magnification. Suitable fields in the preparation were photographed [20]. 16S rRNA gene sequencing with universal primers and phylogeny analysis using type strains were carried out in EzTaxon-e server and MEGA 6.0 [21].

2.3.2. Cultural characterization

The cultural characteristics of the strain was determined in accordance with the method described by Shirling and Gottlieb, 1966 [16]. Cultural characteristics were observed on yeast extract malt extract agar (ISP-2), inorganic salt starch agar (ISP-4), peptone yeast extract iron agar (ISP-6), glycerol asparagine agar (ISP-5) and yeast extract glucose agar. National bureau of standards colour chart was used to determine the colour of the substrate mycelia, aerial mycelia, and spore mass and pigment production.

2.3.3. Biochemical characterization

Various biochemical tests includes IMViC, urease, H_2S production, oxidase, catalase, coagulase and hydrolysis of casein, lipid, starch, gelatin were carried out for the biochemical characterization of active metabolite producing strain.

2.4. Production of anti-MRSA molecule

Shake flask fermentation and Agar Plate Fermentation (APF) was employed to define the adequate fermentation method and APF with solid medium inoculated with a spore suspension of about 10^7 CFU/mL had found to be optimum for the production of metabolites. A well sporulated mycelium of SBRK-2 was inoculated over the broth medium containing soluble starch 20 g, KNO3 1 g, NaCl 0.5 g, K2HPO4 0.5 g, MgSO4 0.5 g, FeSO4 20 mM, distilled water 1 L and incubated at 28 °C up to 14 days in a shaker. After 14 days of incubation cells were harvested and homogenized using micropestle and spore suspension was prepared. 50 μL of the spore suspension containing approximately 1×10^6 CFU/mL of the strain *Streptomyces* sp SBRK-2 was swabbed on agar plates containing AIM medium and incubated for 9 days at 28 °C. At the end of this APF, the grown mycelial cake was cut into small pieces and the organic compounds were extracted using double the volume of methanol for 24 h and the solvent phase was removed. The organic methanolic extract was filtered and concentrated using vacuum concentrator (Eppendorff 5305, Germany). The dried methanolic extract was quantified and Anti –MRSA efficacy was further confirmed with Disc diffusion method [22].

2.5. Bioactivity guided fractionation and spectral analysis

The crude methanolic extract was subjected to a bioactivity guided fractionation. One gram of crude extract was loaded onto a silica gel column using gradient elution with an increasing polarity of chloroform (1–100%) and methanol. Twenty elutions (C1 to C20) were collected with 15 mL for each fraction. All the fractions were concentrated in vacuum concentrator at 30 °C and tested for the active component against MRSA ATCC33591. The column active fractions were pooled together and subjected to Thin Layer Chromatography (TLC) in order to separate the active molecule. The activated TLC silica gel 60 F_{254} sheet (Merck) was used as a stationary phase with the solvent system chloroform: methanol in the ratio of 9:1. Chromatograms were observed under UV light. The R*f* value of each spot was calculated by the formula, R*f* = Distance Analyte travels/Distance solvent travels. Each spot was scrapped separately and dissolved in methanol, left overnight. It was centrifuged at 8000 rpm for 5 min; the supernatant was concentrated and tested for anti –MRSA activity by disc diffusion method. One set of developed TLC plate was dried and kept in dark. 24 h old cultures of MRSA ATCC33591 was grown on LB broth and the culture (10^7 CFU inoculum size) was mixed with soft agar (0.6%) and carefully flooded over the TLC sheet. Thereafter, the plates were incubated overnight at 37 °C and 100% relative humidity in the dark and then sprayed with a 2 mg/mL solution of *p*-iodonitrotetrazolium violet (Sigma®) (INT). White bands indicate where reduction of INT to the coloured formazan did not take place due to the presence of compounds that inhibited the growth of tested organisms [23,24]. TLC purified active fraction (T5) was subjected to reverse phase high-performance liquid chromatography (RP-HPLC) system using C18 column (Sunfire-C18, 4.6 × 250 mm) with Photo Diode Array (PDA) detector (Waters). The mobile phase was applied as the linear gradient. The chromatographic parameters are the following: detection –PDA (200–800 nm); mobile phase: A - Water, B - Methanol; 0–2 min 70% A–30% B, 2–10 min 10% A- 90% B, 10–20 min 50% A-50% B, 20–30 min 70% A-30% B; and the flow rate −1 mL/min. Seven peaks were observed in HPLC analysis using PDA detector. The HPLC purified inhibitory fraction was analyzed by high resolution liquid chromatography-mass spectrometry (HR-LC-MS) with Electro Spray Ionization (ESI) source and Orbitrap mass detector (Thermo Scientific Exactive) at ultra-high resolution in order to determine the exact molecular weight and structure of *anti*-MRSA molecule. The results were detected in the positive ion mode and *m/z* 50–2000 of the scanning mass scan range in full scan MS. The chromatogram peaks of the obtained components in LC-MS were subjected to mass-spectral analysis. The spectra were analyzed from the available library data using Dictionary of Natural products, Knapsack database and NIST MS Search. Molecular vibrational stretches of active molecule was analyzed using FTIR spectra. The KBr pellet (13 mm) prepared using the pure compound was used to obtain the FTIR spectra in the scanning range of 400–4000 cm^{-1} at a resolution of 4 cm^{-1} [25].

2.6. Antibacterial assay

Minimum inhibitory concentration (MIC) was determined according to Clinical and Laboratory Standards Institute (CLSI) guidelines [26]. MRSA colonies were inoculated in 10 mL of cation-adjusted Meuller--Hinton broth with 1% NaCl and incubated for 6–8 h at 37 °C, with shaking at 150 rpm. The turbidity of the bacterial suspension was adjusted with sterile broth to absorbance values of 0.07–0.1 at 620 nm to achieve a suspension containing ~1 × 10^7 CFU/mL. The adjusted inoculum was further diluted 100-fold and used for the antibacterial assay. The inoculum was added into a 96- well flat-bottom microtiter plate containing Mueller-Hinton broth with 2% NaCl and 1% DMSO as a carrier. Different concentration of HPLC purified inhibitor (1 μg/mL,2 μg/mL, 4 μg/mL, 8 μg/mL, 16 μg/mL and 32 μg/mL was added to each well. MRSA treated with vancomycin and vehicle solvent (1% DMSO) was used as positive and growth controls, respectively. The final density of the bacteria per well was ~5 × 10^5 CFU/mL. The plate was sealed and incubated with shaking for 16–20 h at 37 °C. The growth turbidity of bacterial suspension was measured using multimode plate reader (PerkinElmer) at 620 nm. The amount of growth in each well is compared with the positive growth control and the MIC recorded as the lowest concentration of the inhibitor that completely inhibits growth of MRSA [27]. After MIC determination, aliquots of 100 μL from each well, which showed no bacterial growth after incubation was seeded in Mueller-Hinton agar plates. The lowest concentration of the *anti*-MRSA molecule that kills 100% of the initial bacterial population showing no colony on the MH agar after 24 h of incubation at 37 °C was recorded as the MBC [28].

2.7. Time-kill kinetics

Time-kill kinetics of *anti*-MRSA molecule was carried out following the procedure described by Theresa et al. [29]. 1/2x MIC, 1x MIC, 2x MIC and 4x MIC concentrations of the *anti*-MRSA molecule was prepared. An inoculum size of 1.0 × 10^6 CFU/mL was added and incubated at 37 °C. Aliquots of 1.0 mL of the medium were taken at time intervals of 0, 2, 6, 12 and 24 h, for bacteria and inoculated aseptically into 20 mL Luria Bertani agar and incubated at 37 °C for 24 h. A control test was performed for the organisms without *anti*-MRSA molecule. The colony forming unit (CFU) of the organisms was determined. The procedure was performed in triplicate (three independent experiments) and a graph of the log CFU/mL was plotted against time. Bactericidal activity was defined as a decline of >30 CFU/mL (99.9%) killing in bacterial density from the starting inoculum.

2.8. Evaluation of anti-biofilm activity

Biofilm inhibition effect of anti–MRSA compound was performed using static biofilm model and biofilm formation was determined by the crystal violet staining method [30]. In the 96-well polystyrene plate, 1% DMSO and various concentrations of *anti*-MRSA agent (1/4 × MIC, 1/2 × MIC, MIC, 2 × MIC and 4 × MIC of anti MRSA molecule and 2 × MIC of vancomycin) was added. The plates were incubated under aerobic conditions at 37 °C for 48 h. Liquid mixture was discarded and wells were stained with 0.1 mL 0.4% crystal violet for 15 min after being washed twice with sterile water. Afterwards, the dye bound to biofilm was solubilized by adding ethanol (95%). Absorbance of the isolated dye was measured quantitatively at 540 nm. The control was maintained without inhibitor. The biofilm inhibition percentage was calculated using the following formula.

[(OD growth control−OD sample)/OD growth control] × 100.

In order to test the effects of *anti*-MRSA compound on the biofilm architecture and formation of biofilm Epi-Fluorescence Microscopy was used. The formation of the biofilm was made in 1 cm × 1 cm glass coverslips using an inoculum of MRSA ATCC33591 in saline solution at OD_{600}nm 0.05 in liquid LB medium containing the biofilm-inhibitor compounds. Thus, coverslips were incubated at 37 °C for 24 h. The fixation and drying of the material were done by heating at 37 °C for 12 h [31]. One set of coverslips were rinsed with water and then immersed in 0.1% acridine orange in phosphate buffered saline pH = 7.2 (PBS) for 1 min; the excess dye was rinsed off with sterile water. The stained biofilm was observed with an Epi-Fluorescence attachment, using a camera and the fluorescence intensity was measured (NIS-ELEMENTS).

2.9. Influence of anti-MRSA agent on cell-surface hydrophobicity

Bacterial adherence to hydrocarbon (BATH) assay was performed to measure *S. aureus* cell hydrophobicity. Briefly, 100 μL of *S. aureus* ATCC 33591 cell suspension containing 5 × 10^5 CFU/mL was cultivated in 900 μL of tryptic soy broth overnight. The broth were centrifuged at 4000 g for 5 min and the pellet containing cells were washed and resuspended in 2 mL of 0.9% sodium chloride supplemented with the 0 (negative control), 1/4 × MIC, 1/2 × MIC, MIC, 2 × MIC and 4 × MIC of antibacterial compound and incubated at 37 °C for 24 h 2× MIC of vancomycin was used as control antibiotic. Toluene (250 μL) was treated to each cell suspension as described earlier by Iniyan et al. [32], and the toluene phase was separated from the aqueous phase and the absorbance of the aqueous phase was taken at 600 nm. The hydrophobicity index (HPBI) was calculated as: (OD initial-OD final/OD initial) x 100%. The changes in cell surface hydrophobicity of treated cells were calculated.

2.10. Cell surface analysis by SEM and AFM

Overnight grown MRSA ATCC 33591 cells were harvested and pretreated with the compound at its $1/2 \times$ MIC concentration. The compound treated with bacterial cells was incubated for 1 h at 37 °C to find the morphological changes. Controls were prepared without compound treatment. The cells were then air dried and coated with gold and examined under the scanning electron microscope as described previously. The effect of compound on MRSA ATCC 33591 cell envelope was further investigated by atomic force microscopy analysis. The atomic force microscopy images of the cells treated with $1/2 \times$ MIC concentration for 1 h were analyzed by NOVA software in order to visualize the effect of the treatment and to determine the values of cell surface roughness.

2.11. Hemolysis assay

Hemolytic activity of the *anti*-MRSA compound was analyzed in human red blood cells. Red blood cells were rinsed several times in PBS by centrifugation for 3 min at 3000 *g* until the OD of the supernatant reached the OD of the control which has PBS alone. Red blood cells were counted using hemocytometer and adjusted to $7.7 \times 10^6 \pm 0.3 \times 10^6$ cells/mL. Red blood cells were then incubated at room temperature for 1 h in 10% Triton X-100 (positive control), in PBS (blank), or with anti-MRSA compound at concentrations of $1/4 \times$ MIC, $1/2 \times$ MIC, MIC, $2 \times$ MIC and $4 \times$ MIC. The samples were then centrifuged at 10,000 *g* for 5 min, the supernatant was separated from the pellet, and its absorbance measured at 570 nm. The relative optical density compared to that of the suspension treated with 10% Triton X-100 was defined as the percentage of hemolysis.

2.12. Safety assessment in zebrafish embryos

Zebrafish were maintained in active-green stand-alone zebrafish system. Following successful breeding, eggs were collected from the breeding tanks and maintained according to Wester field 1989. The chemical genetic effect was studied by observing the phenotypic deformities of the *anti*-MRSA molecule treated zebrafish embryo. The observations included pigmentation, size of the embryo, changes in yolk sac, deformities in eyes, brain, heart, tail and trunk. The embryos were anesthetized by 0.01% tricane (Sigma) and the observations were carried out under a stereo fluorescence microscope (Leica M165 FC).

3. Results

3.1. Identification of active strain

Based on the morphological and 16S rRNA gene sequences the *anti*-MRSA producing actinomycete strain was identified as *Streptomyces* sp. SBRK-2 (Genebank Acession Number: JX100418.1) which is closely related to *Streptomyces longispororuber* NBRC 13488^T. The growth on the isolation media showed pinkish grey aerial mycelial colour and grey substrate mycelia pattern (Fig. 1a). The spore morphology of the strain was observed under SEM and found to be smooth spore surface (Fig. 1b). 16S rRNA gene sequencing analysis showed that the *anti*-MRSA

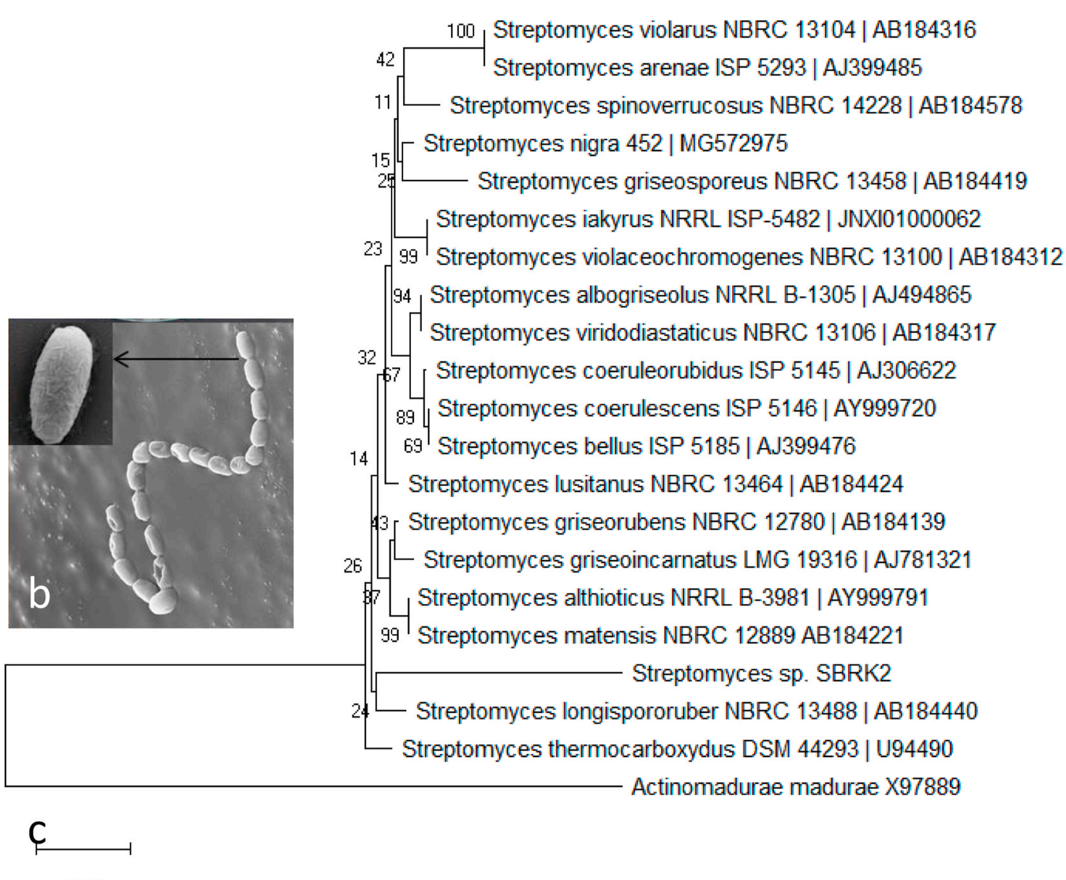

Fig. 1. *Streptomyces* sp. SBRK2 morphology (a), scanning electron microscopy image of spore surface (b) and neighbour joining phylogenetic tree using 16S rRNA gene sequences (c).

compound producer *Streptomyces* sp. SBRK-2 was closely related to *Streptomyces coerulescens* ISP 5146^T in the EzTaxon type strain database. The neighbour joining phylogenetic tree was shown in Fig. 1c. Characterization of the strain SBRK2 showed aerobic, Gram-positive forms distinct aerial and substrate mycelial pattern on ISP media. Colours of the aerial mycelia were observed as pinkish white to grey combination. On peptone yeast extract/iron agar or tyrosine agar media it produces dark brown diffusible pigment and also possess earthy odour. The growth pattern was depicted in Table 1. Biochemical analysis of SBRK2 showed positive results in voges proskeur, catalase and citrate utilization test. The strain was able to hydrolyse starch and casein in the medium and produce H_2S on Triple Sugar Iron (TSI) media (Table 2).

3.2. Isolation of antibacterial active compound

The crude methanolic extract was initially purified by column chromatography and separated the active fractions at elution numbers 12, 13 and 14 (Fig. 2b). They were pooled together and further separated by TLC in silica gel 60 F_{254} sheets. Totally seven spots were resolved and the R*f* value of each spots were determined and shown in Fig. 3a. Disc diffusion assay revealed that the fraction with R*f* value of 0.57 (T5) showed *anti*-MRSA activity (Fig. 2c). Likewise the TLC plates were subjected to the direct bioautography in order to detect the *anti*-MRSA spot on the developed TLC sheet. The INT dye based detection revealed the zone of inhibition in the region due to the absence of live cells in the background of pink colour formation of reduced INT (Fig. 3a). The TLC active fraction T5 was purified in analytical RP-HPLC in which the active fraction was detected for the peak with the retention time of 14.86 min at 285 nm absorbance in the PDA detector (Fig. 3b). Further the purity was checked in analytical mode of HPLC (Fig. 3c) and the active fraction was characterized using LC-HRMS. LC-ESI-MS analysis of the HPLC purified peak eluted in the chromatographic peaks at tR = 2.36 min, exhibiting the molecular mass deduced by the ion at *m/z* 338.3421 and 360.3241 corresponding to corresponding to $[M+H]^+$ and $[M+Na]^+$, respectively, in positive ion spectrum (Fig. 3d). The molecular weight of the compound was identified as 337.3342 Da. Dereplication analysis of the mass identified the compound as 8-*O*-metyltetrangomycin with the molecular formula of $C_{20}H_{16}O_5$. Molecular vibrational stretches of the molecule was analyzed using FTIR analysis which has correlated with the structure of 8-*O*-metyltetrangomycin. FTIR frequencies showed five major peaks as 3083.89, 2862.36, 1651.18, 1260.66 and 897.66 cm-1. The peak at 3083.89 cm-1 from the molecule spectra is most likely due to C=C-H asymmetric stretch of alkenes and aromatic benzene ring. A peak at 2862.36 cm-1 is most likely due to the presence of H-C-H asymmetric and symmetric stretch and O-H stretch of carboxylic acids. A peak at 1651.18 is due to the presence of C=O stretch of ketones.

3.3. Anti-MRSA activity

Streptomyces sp. SBRK2 was found to exhibit strong *anti*-MRSA activity with 28 mm diameter zone of inhibition (Fig. 2a) and the crude methanolic extract showed maximum inhibition in disc diffusion assay.

Table 1
Physiological characterization *Streptomyces* sp SBRK2 on different ISP media.

ISP	Aerial mycelium		Substrate mycelium	
ISP1	++	Greyish pink	++	Grey
ISP2	+	Greyish white	+	Grey
ISP3	++	Dark grey	++	white
ISP4	–	No aerial	++	Grey
ISP5	+++	Whitish grey	++	Whitish grey
PEPTONE IRON AGAR	+++	Dark grey	+++	Brownish grey
TRYPTONE IRON AGAR	++	Greyish orange	+	Dark brown

Note: = Absent, + = present, ++ = moderate; +++ = High.
ISP = International Streptomyces Project.

Table 2
Biochemical analysis of beta lactamase producing strain *Streptomyces* sp SBRK2.

Test	Result	Test	Result
Gram's Staining	+	Catalase	+
Indole	–	Citrate Utilization	+
Methyl red	–	coagulase	–
Voges Proskauer	+	Casein Hydrolysis	+
Urease	–	Lipid Hydrolysis	–
H_2S production	+	Starch Hydrolysis	+
oxidase	–	Gelatin Hydrolysis	–

Note: =Negative; + = positive.

The active compound showed antibacterial activity against MRSA strain and its MIC against MRSA was identified as 2 μg/mL while the MBC that is minimum concentration required to kill bacteria was 4 μg/mL. The time-kill kinetics profile of the compound 8-*O*-metyltetrangomycin against MRSA showed reduction in number of viable cells with increase of time. MIC of the compound demonstrated a $\geq$3.5-$\log_{10}$-unit killing of the final inoculum. The time-kill curves for 2x MIC and vancomycin looked similar with almost rapid killing of bacteria within the assay time of 24 h 4x MIC demonstrated the greatest killing effect at 12 h of incubation with a $\geq 1\log_{10}$ unit killing (Fig. 2d).

3.4. Cell surface analysis and biofilm inhibition assay

In SEM analysis, the micrographs revealed that untreated MRSA appeared as cocci, grape-like cluster or chain, smooth surface and plump appearance. The bacterial membrane layer was disrupted upon treatment with 1x MIC of 8-*O*-metyltetrangomycin (Fig. 4). Bacterial cell membrane was severely disrupted with evident plasmolysis upon treatment. Massive leakage of cell content was observed in all treated cells. Membrane damaging activity, where they would target on the hydrophobic bacterial surface, modifies the membrane fluidity and disrupts the cell membrane structure.

Atomic force microscopy was used to visualize the effect of 8-*O*-metyltetrangomycin on bacterial cells. The untreated MRSA cells were observed to be cocci shaped (Fig. 5a). After 8-*O*-metyltetrangomycin treatment, the MRSA cells were severely damaged and possibly dissolved (Fig. 5a1). The 3D image and line profile of MRSA treated cells clearly indicated the cellular damage and change in size and morphology of the cell when compared with untreated cells (Fig. 5b, c, b1 and c1). Atomic force microscopy results indicated that 8-*O*-metyltetrangomycin interact directly with cell membrane components, resulting in bacterial cell death.

Bacterial biofilm inhibition of *anti*-MRSA compound was tested and percent inhibition was calculated. Dose dependent inhibition of biofilm formation was observed and it showed significant antibiofilm activity ranged from 52.85% to 86.64% inhibition (Fig. 6E). The MIC fraction of 8-*O*-metyltetrangomycin exhibited more antibiofilm potential than vancomycin and highest range of inhibition was observed at 4x MIC of the compound, it suggests that the compound possess stronger potential to reduce biofilm formation. In coverslips the biofilm formed by MRSA without any treatment shown as dense groups and the fluorescence intensity also more prominent, the green peaks indicates the viable cells in active biofilm. At the same time, bacterial biofilm density and viability was notably decreased in treated samples. Also, bacterial clusters, although spread throughout the surface and forms thin layers, which is evidenced by the reduction in the light intensity of the staining is depicted in Fig. 6A,A1, B,B1, C, C1 and D, D1. The effect of sub-MIC concentrations of 8-*O*-metyltetrangomycin on cell-surface hydrophobicity is depicted in Fig. 6E. The BATH assay on *S. aureus* ATCC 25923 cells revealed significant increases on cell surface hydrophobicity from the initial concentration with one fourth MIC concentration of 8-*O*-metyltetrangomycin exposure. At ½ X MIC concentration 74.77 ± 2.79% hydrophobicity was observed and the control showed 18.41 ± 2.79% hydrophobicity which is statistically significant ($P < 0.05$).

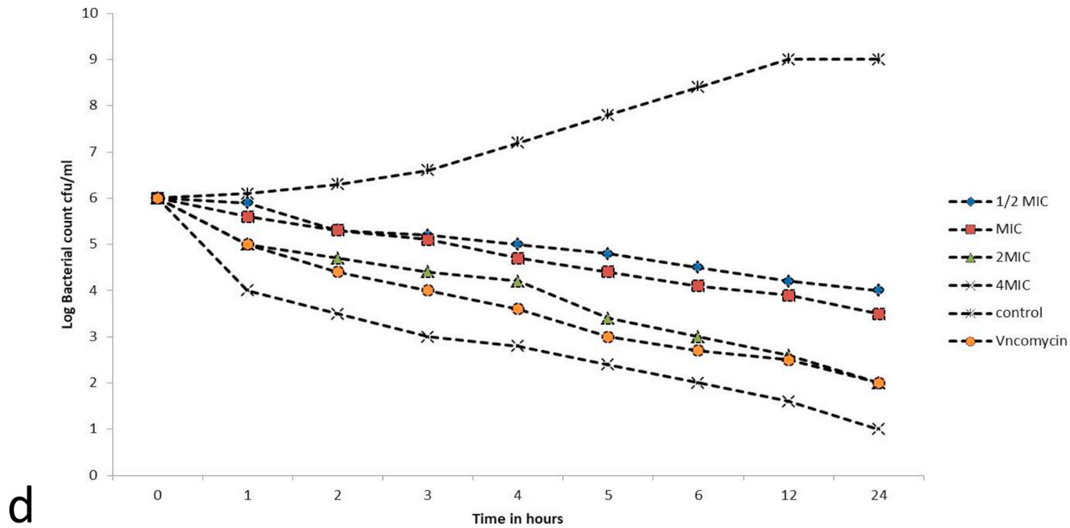

Fig. 2. *Anti*-MRSA activity by double layer assay (a) and column fractions in disc diffusion assay (b and c). Time-kill curves for *anti*-MRSA with four different concentrations and vancomycin as positive control.

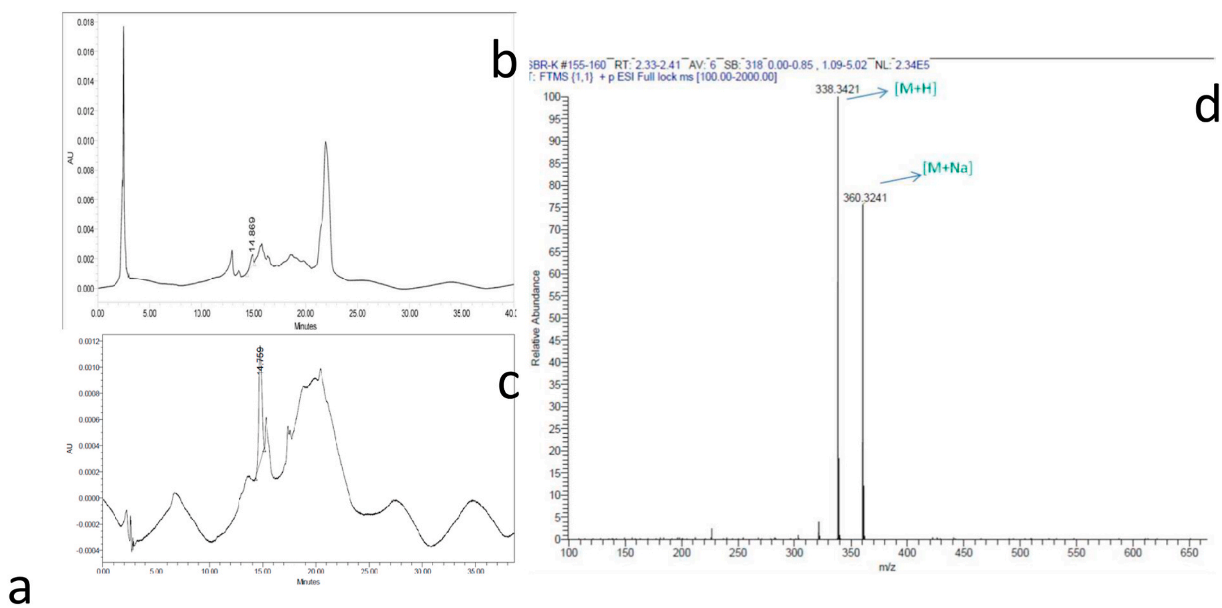

Fig. 3. Purification profile of *anti*-MRSA compound by TLC bioautography (a) HPLC (b,c) and LC-MS chromatogram. The analytical HPLC showed a compound at the retention time of 14.769 min is the target compound. Other peaks are the baseline drift and confirmed in HPLC with the Methanol and Water in the quadruple gradient system. Mass spectrum of retention time 2.36 min shows the mass of the compound 337.3342 Da. The molecular ion peak shows the *m/z* 338.3421 and 360.3241 corresponding to corresponding to [M+H]+and [M+Na]$^+$.

3.5. Hemolysis assay and toxicity assessment in zebrafish embryos

The hemolytic assays of *anti*-MRSA compound 8-*O*-metyltetrangomycin on human erythrocytes showed that the 2 × and 4 × MIC concentrations had hemolytic activities with 41% and 63%, respectively. Pharmacological studies in zebrafish embryo revealed 8-*O*-metyltetrangomycin was found to be safe up to 100 μg/mL which is 50 times higher concentration than its MIC value. There has no significant changes in heart beat rate was observed up to 100 μg/mL and the higher assay concentration at 150 μg/mL resulted in reduced HBR as 142.6 ± 2.37 beats/min. The LC_{50} of the compound was found to be 146.67 μg/mL and the 100 mg/mL was found to be maximal non-lethal concentration.

4. Discussion

The present study reports the isolation and antibiofilm potential of an angucycline class of antibiotic 8-*O*-metyltetrangomycin from a marine sponge associated *Streptomyces*. The ever increasing bacterial resistance to antibiotics and the toxicity of many antibiotic compounds necessitate active efforts on antibiotics in the hope of finding new efficient and less toxic compounds in order to control drug resistant microbes [33–35]. Our previous works have already established the biodiversity and richness of actinomycetes in the coastal sediments, marine sponges and mangroves of Southern coast of India. These studies have led to the discovery of novel antibiotic Ala-geninthiocin and interesting bioactive compounds [13,36,37] and a new species of actinobacteria [38]. The actinobacterium strain SBRK2 was isolated from a sponge of Gulf of Mannar, Rameswaram, India has produced

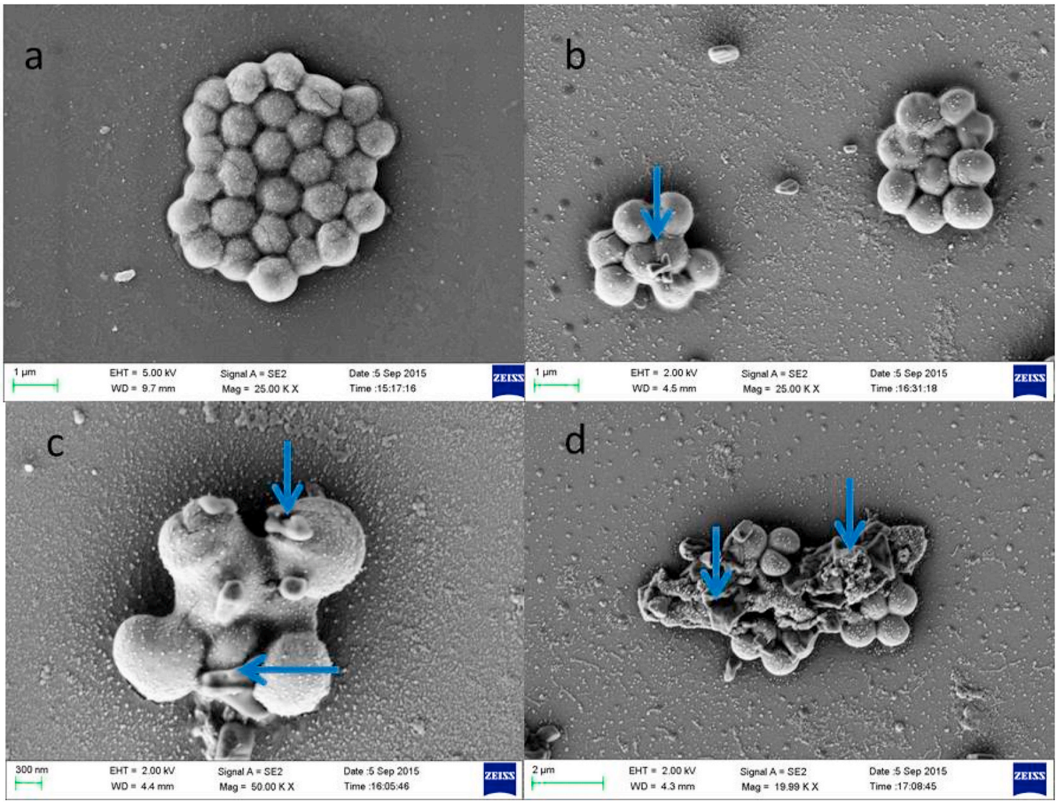

Fig. 4. Scanning electron microscopy (SEM) photograph of MRSA treated with *anti*-MRSA compound 8-*O*-metyltetrangomycin. (a) Control (b) treated with ½ X MIC, (c) treated with MIC and (d) treated with 2× MIC of the isolated compound. The 2× MIC treated cells were disrupted severely and plasmolysis occurred which is shown in blue arrows. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

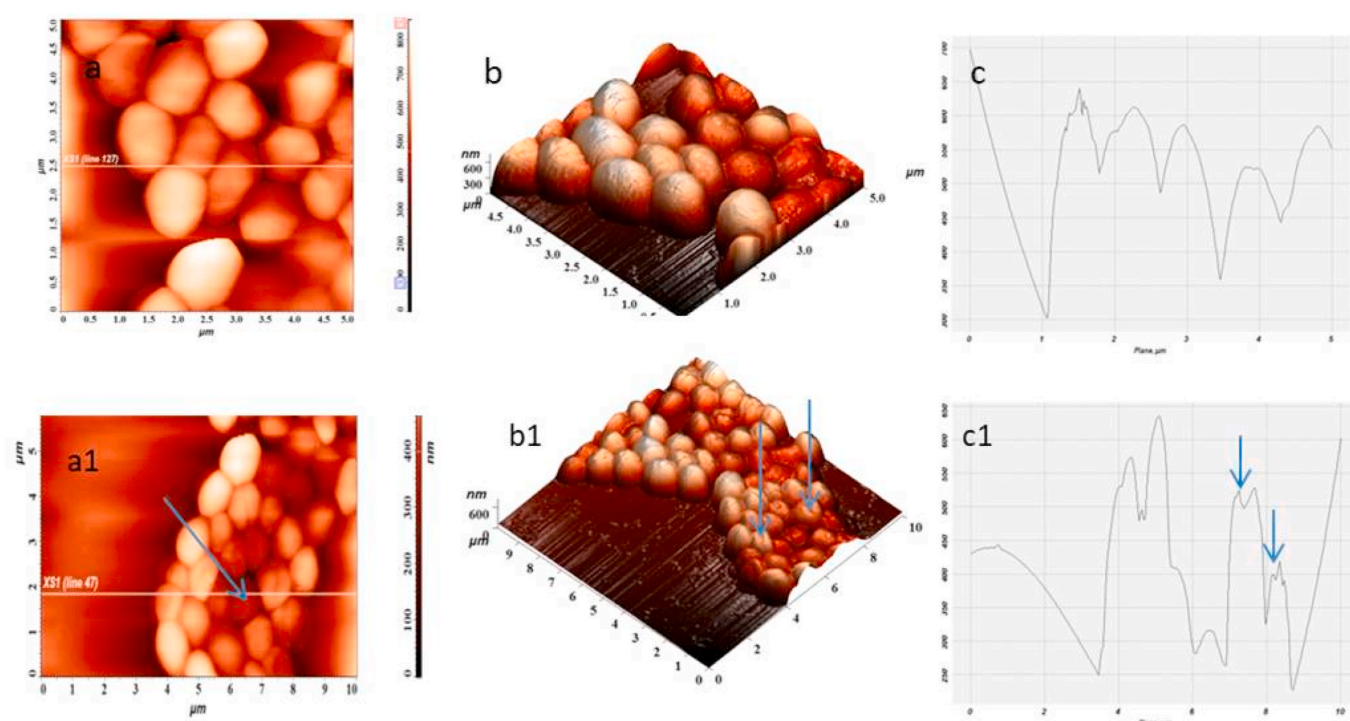

Fig. 5. AFM of 8-*O*-metyltetrangomycin treated MRSA cells. (a, b and c) represent AFM images of untreated, MRSA, cells (a1,b1 and c1) represent AFM images of MIC (2 µg/mL) treated MRSA cells. Blue arrow indicates the cellular damage. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

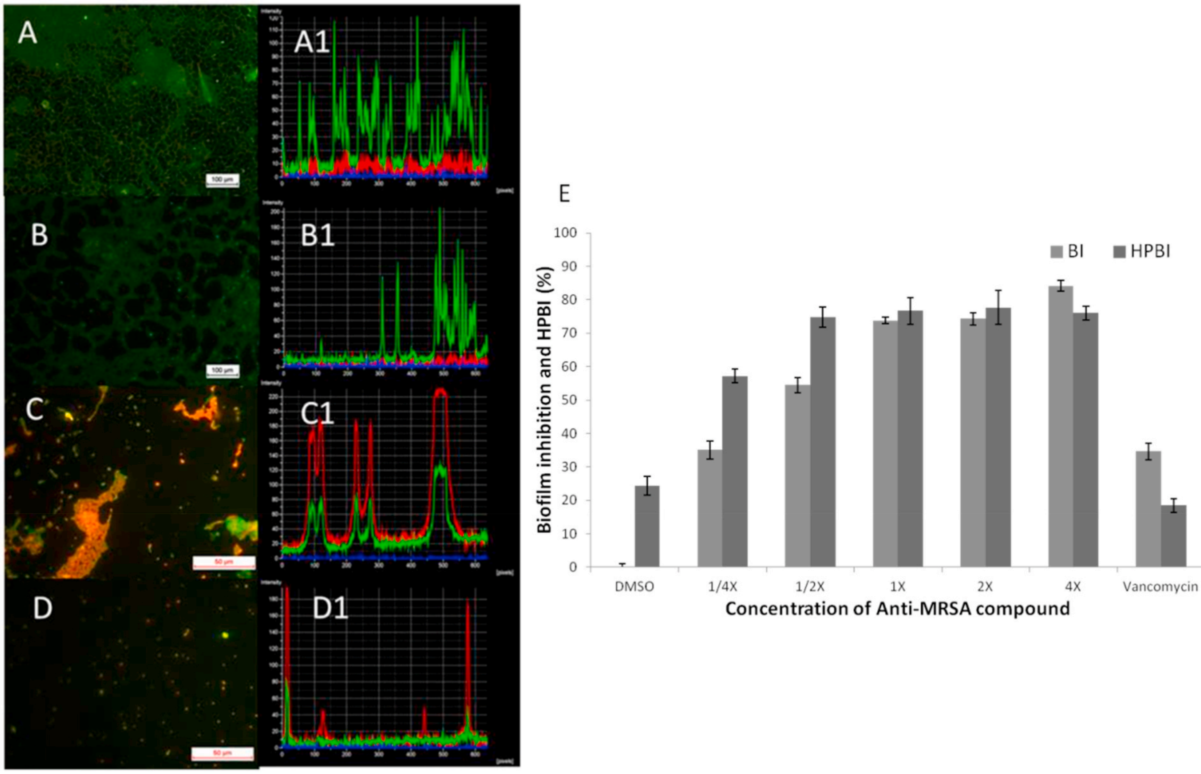

Fig. 6. Biofilm inhibitory potential of 8-*O*-metyltetrangomycin from *Streptomyces* sp SBRK2 at different MIC levels on coverslips against *S. aureus* ATCC 25923. All biofilms were stained with acridine orange at 0.1% (A to D) and the fluorescence intensity was measured (A1 to D1). E shows the effect of 8-*O*-metyltetrangomycin on cell-surface hydrophobicity and quantitative biofilm inhibition. One way ANOVA revealed that the changes in cell-surface hydrophobicity and biofilm inhibition were statistically significant ($P < 0.05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

8-*O*-metyltetrangomycin. The angucycline group metabolites have been shown to have a variety of biological activities, such as antibacterial and antitumor activity, and some act as inhibitors of oxidative enzymes. Many of these anthraquinone derivatives so far isolated were weakly active against Gram positive and Gram negative bacteria. EI-1507-1 inhibited growth of *Enterococcus faecium, Bacillus subtilis* and *Proteus vulgaris* with MIC values of 120, 60 and 60 mM, respectively [39]. MM 47755, an antibiotic from a graysporing Streptomycete isolated from soil exhibit antimicrobial activity against Gram-positive bacteria *Bacillus subtilis* with MIC of 32 μg/mL.

The antibacterial effect against MRSA is comparable to the previously reported antimicrobials with a pyrrole-like structure, Moenomycin A [40], nosokomycins A- D and angumicynone B [41] which also were isolated from marine *Streptomyces* and reported as potential marine drugs against MRSA. The observed anti- MRSA MIC of the compound from *Streptomyces* sp. SBRK2 (2 μg/mL) was 6 fold lower than that of angumicynone B (12.5 μg/mL) and 2 fold lower than moenomycin A (4 μg/mL), and higher than nosokomycins A- D 0.125 μg/mL [42]. Moreover, this MIC is lower than vancomycin (4–8 μg/mL) which is a commercially available *anti*-MRSA antibiotic. Thus, the observed MIC and MBC of the isolated compounds in our study, demonstrate that the compound exhibited a significant antimicrobial activity against MRSA and it may exhibit effective antibacterial activity against other drug- or multidrug-resistant bacteria especially other *Staphylococcus* and related genus.

The mechanism of antibacterial activity has a major role for each class. Ampicillin and vancomycin could inhibit bacterial cell wall synthesis. It was reported that the potency of ampicillin and vancomycin can be enhanced if they were administered together with the phenolic acids, in a nonspecific manner [43]. Amphiphilic nature of phytochemicals is closely associated to the molecular interaction with the cell, attachment to cell membrane and penetration into the cell. Chen et al. reported earlier the SEM analysis showed the cell membrane of MRSA were destroyed after treating actinomycin ×2 and collismycin A [44]. Our angucycline compound 8-*O*-metyltetrangomycin affects the cell surface was identified using SEM and AFM analysis. Recently, Yang et al. [45] reported arginine-rich α helical antimicrobial peptide SRP-2 exhibits direct membrane disruptions of the bacterial cells including *E. coli* and MRSA.

Staphylococcus aureus is responsible for several community-acquired and clinical infections. Quorum sensing is a process in which chemical signals produced by microbes itself help in the organization of biofilm into complex structures [46]. Moreover, it is important for the production of virulence factors like biofilm, acid tolerance, and staphyloxanthin biosynthesis, etc. in *S. aureus. Staphylococcus aureus* produces golden pigment i.e. staphyloxanthin, a major virulence determinant, functioning as a redox active toxin and also promote release of extracellular DNA which help in the biofilm formation [47]. Hence, the antibiofilm potential of the isolated compound 8-*O*-metyltetrangomycin was evaluated and found the sub-MIC concentrations could inhibit biofilm formation which was also supported by the hydrophobicity index. Selected angucyclines are also potent inhibitors of blood platelet aggregation [48]. Similarly, 8-*O*-metyltetrangomycin showed haemolytic activity in red blood cells. Toxicity assessment in zebrafish revealed the compound is safe which could be an added advantage to explore the potential of this compound.

In conclusion, the strain SBRK2, related to *Streptomyces longispororuber*, produced this *anti*-MRSA compound 8-*O*-metyltetrangomycin. This angucycline family antibiotic showed *anti*-MRSA activity with antibiofilm potential. Considering the important biological activities of angucyclines, it would be interesting to explore the broad spectrum antimicrobial activity of 8-*O*-metyltetrangomycin and other properties of these new angucyclines molecule including anticancer activity which could prove promising.

Author statement

Thankaraj Rajam Jabila Mary- Methodology, Validation, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Visualization

Rajaretinam Rajesh Kannan – Conceptualization, Validation, Investigation, Resources, Data Curation, Writing - Review & Editing, Visualization, Supervision, Project administration, Funding acquisition.

Appadurai Muthamil Iniyan- Formal analysis, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing

Dasnamoorthy Ramachandran - Formal analysis, Visualization

Samuel Gnana Prakash Vincent- Conceptualization, Resources.

Declaration of competing interest

The authors declare that they have no conflict of interest.

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